

Lab 07: Methods I

Objective

- To practically familiarize student with how to write the common code once and reuse it without rewriting it using the concept of Methods.

Current Lab Learning Outcomes (LLO)

By completion of the lab the students should be able to

1. define their own methods
2. invoke (call) a method (method invocation)
3. passing parameters by values
4. how a method may return a value

Lab Requirements

NetBeans IDE 8.0.1 or higher

Lab Assessment/Activities

Practice Activity with Lab Instructor (10 minutes)

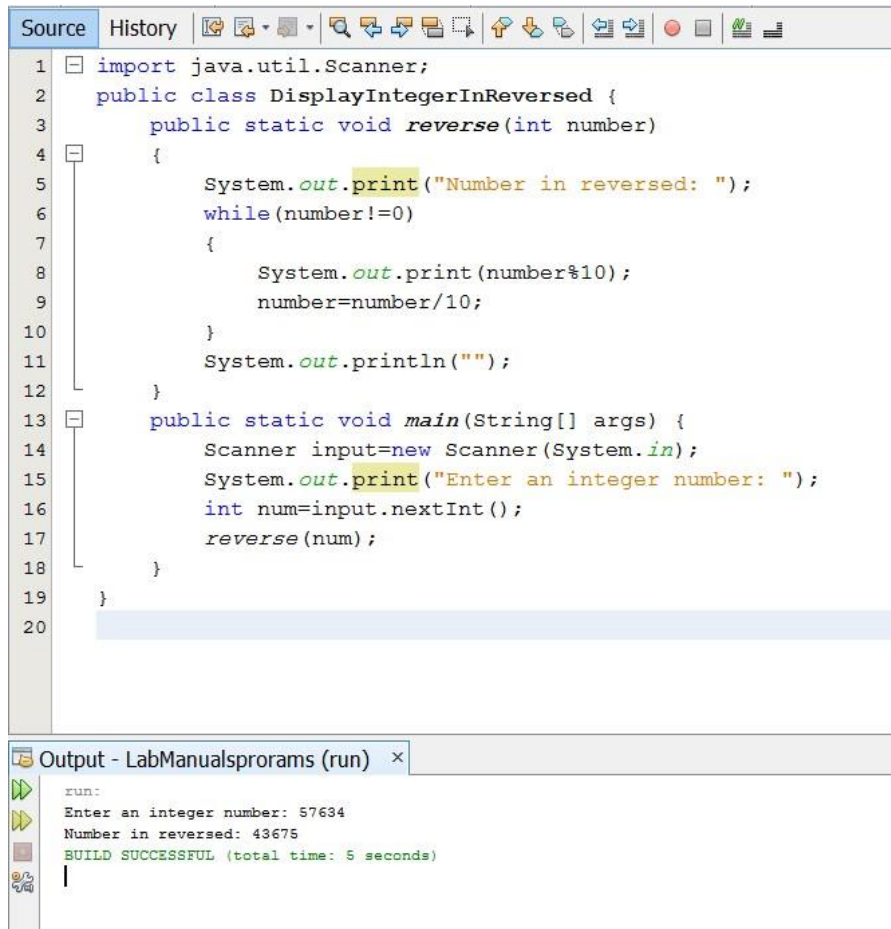
(Display an integer reversed) Write a method with the following header to display an integer in reverse order:

```
public static void reverse(int number)
```

For example, `reverse(3456)` displays `6543`. Write a test program that prompts the user to enter an integer and displays its reversal.

Solution:

1. Open NetBeans and create a new project
2. Create a new java main class and write its name as **DisplayIntegerInReversed**
3. Write the following code inside the main method



```
1 import java.util.Scanner;
2 public class DisplayIntegerInReversed {
3     public static void reverse(int number)
4     {
5         System.out.print("Number in reversed: ");
6         while(number!=0)
7         {
8             System.out.print(number%10);
9             number=number/10;
10        }
11        System.out.println("");
12    }
13    public static void main(String[] args) {
14        Scanner input=new Scanner(System.in);
15        System.out.print("Enter an integer number: ");
16        int num=input.nextInt();
17        reverse(num);
18    }
19 }
20
```

Output - LabManualsprorams (run) x

```
run:
Enter an integer number: 57634
Number in reversed: 43675
BUILD SUCCESSFUL (total time: 5 seconds)
```

Short Exercise (5 minutes)

(Sum the digits in an integer) Write a method that computes the sum of the digits in an integer. Use the following method header:

```
public static int sumDigits(long n)
```

For example, `sumDigits (234)` returns `9` ($2 + 3 + 4$).

Individual Activities: (60 minutes)

1. **(Count the letters in a string)** Write a method that counts the number of letters in a string using the following header:

```
public static int countLetters(String s)
```

Write a test program that prompts the user to enter a string and displays the number of letters in the string.

2. **(Math: pentagonal numbers)** A pentagonal number is defined as $n(3n-1)/2$ for $n = 1, 2, \dots$, and so on. Therefore, the first few numbers are 1, 5, 12, 22, $\dots$. Write a method with the following header that returns a pentagonal number:

```
public static int getPentagonalNumber(int n)
```

Write a test program that uses this method to display the first 100 pentagonal numbers with 10 numbers on each line.

3. **(Conversions between Celsius and Fahrenheit)** Write a class that contains the following two methods:

```
/** Convert from Celsius to Fahrenheit */
```

```
public static double celsiusToFahrenheit(double celsius)
```

```
/** Convert from Fahrenheit to Celsius */
```

```
public static double fahrenheitToCelsius(double fahrenheit)
```

The formula for the conversion is:

$$\text{fahrenheit} = (9.0 / 5) * \text{celsius} + 32$$

$$\text{celsius} = (5.0 / 9) * (\text{fahrenheit} - 32)$$

Write a test program that invokes these methods to display the following tables:

Celsius	Fahrenheit		Fahrenheit	Celsius
40.0	104.0		120.0	48.89
39.0	102.2		110.0	43.33
...				
32.0	89.6		40.0	4.44
31.0	87.8		30.0	-1.11

4. **(Display matrix of 0s and 1s)** Write a method that displays an n -by- n matrix using the following header:

```
public static void printMatrix(int n)
```

Each element is 0 or 1, which is generated randomly. Write a test program that prompts the user to enter n and displays an n -by- n matrix. Here is a sample run:

```
Enter n: 3
0 1 0
0 0 0
1 1 1
```

5. **(Check password)** Some websites impose certain rules for passwords. Write a method that checks whether a string is a valid password. Suppose the password rules are as follows:
- A password must have at least eight characters.
 - A password consists of only letters and digits.
 - A password must contain at least two digits.

Write a program that prompts the user to enter a password and displays **Valid Password** if the rules are followed or **Invalid Password** otherwise.

Lab Description

- A method is a collection of statements grouped together to perform an operation.
- There are two types of methods: built in methods and user defined methods
 - built in methods are defined in the Java library
 - user defined methods create by user
- Some example for built in methods

System.out.println	System.exit
Math.pow	Math.random
input.nextInt	"some string".toUpperCase

- Defining a method (methods parts, syntax, sample method and method invocation)

A method consists of:

- method name
- parameters
- return value type
- Body

Syntax:

```
modifier returnType methodName(list of parameters) {  
    // Method body  
}
```

Sample method:

- Given two integers, determine the maximum value

Define a method	Invoke a method
<pre>public static int max(int num1, int num2) { int result; if (num1 > num2) result = num1; else result = num2; return result; }</pre>	<pre>int z = max(x, y);</pre>

Extra (supplementary) Materials (if any)

None